# Financial, Managerial, and Valuation Analysis of Edwards Lifesciences

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## BACKGROUND

Edwards LifeSciences (ticker EW) has been around for 64 years. The company was founded by Miles Lowell Edwards, who was the scientist behind the first artificial heart valve ever created in the early 1960s – this was the birth of the industry. Initially, it was called Edwards Laboratories, which was acquired by Baxter in 1985 and eventually spun off by Baxter in 2000 as a publicly traded corporation. At the time, it was the worst performing business unit of Baxter International (which is why it got spun off). Nowadays of course it is a hugely successful business with a rich history.

Edwards sells structural heart solutions from four basic product lines: transcather heart valve replacement (TAVR), transcather mitral and tricuspid therapies (TMTT), surgical heart valve recovery, and critical care devices. The medical device industry is very competitive and extremely dependent on innovative solutions and regulatory approvals. The competition is a combination of much larger corporations who have medical divisions, and smaller companies focused on specific product lines or geographies. Edwards stays competitive by staying extremely focused on their market and providing the more patient value than what their competitors are offering. The main challenges of this business are to remain relevant in an environment with a lot of regulatory approvals and years of clinical data required before products can be released. Edwards must keep good relationships with suppliers as well as protect their proprietary information as much as possible. Typically, sales in the third quarter (Jun-Sept) are the lowest due to people going on vacations in the summer in both US and EU markets - where the majority of sales come from.

Cardiovascular heart disease is the number one cause of death in the entire world, and will always be present in human beings. Therefore, there is always a market for surgical/transcather repair. Additionally, as more patients get repair through Edwards solutions, their reputation grows further, making Edwards a premium solution for all patients struggling with cardiovascular disease. Growth can continue at a strong rate for the foreseeable future, as long as Edwards continues to build its competitive advantage and stay relevant in the space by being first to market on a lot of their products. Additionally, as the baby boomer generation continues to age, Edwards should see an increase in sales given that most of their patients are older people.

Growth Plan (per the 2021 annual report): Edwards growth is through direct sales mainly to customers as well as independent distributors both in the US (57%) and international (43%).

Company culture: Edwards has built an incredible culture of emphasizing work-life balance, diversity, and career building that is unseen in the vast majority of businesses. They have invested heavily into their manufacturing and headquarter locations, where there are a number of athletic facilities including a soccer field, volleyball and pickleball courts, a full service gym, and an incredible dining cafeteria with good healthy food options for incredibly low prices. Employees are given a large amount of autonomy while also allowing growth and learning. Senior leaders are open to mentoring young engineers, and different business units are very open to bringing on team members across different parts of the business in order to help employees find better fits and grow their careers.

## ANALYSIS OF COMPETITIVE ADVANTAGE

Is their product(s) unique and almost essential in modern society? → Solutions to cardiovascular disease are certainly essential in modern society because there have not been proven methods to solve heart disease until now. Most people want to live longer, and thus if they struggle with heart disease, they will need a product from Edwards to survive.

Edwards does innovate constantly but they do so in order to stay dominant in their market. They could just sell their SAPIEN valves and still make a killing in the medical device industry. Once certain products get regulatory approval, they get sold for decades with little additional development required. It's these products that produce consistent revenue for the business. Edwards could lose their competitive advantage if they lose their touch in the market. If they cannot innovate, or are not offering value to patients, they can lose their advantage. However, at the moment Edwards is profiting from three main moats:

1. A strong brand moat. Being around for over 60 years has its benefits. Physicians, doctors, and patients alike have grown used to the name Edwards Lifesciences. Even engineers know the name Edwards and want to go work there because they know they are working on interesting things all the time.
2. Intellectual property. Being in a fiercely competitive industry as they are, Edwards must protect their engineering and scientific achievements with IP. They have over 1,500 current patents as well as 800 pending. This does not include the trade secrets either. (In comparison Medtronic has over 86,000 and Abbott has around 48,000).
3. Network of physicians and doctors. This moat is less noticeable but it is massively advantageous. There is a huge network of physicians who work and are trained specifically with Edwards products. Often times physicians will prefer to work with one company's products over the other, so once a physician is 'switched' then they become a huge asset for the company.

It's good for a company to have a moat, but it's great for a company to have several, and Edwards has three out of the five that are most common (a pricing moat and toll bridge moat are the other two common ones).

Additionally, there is tremendous market potential still available in this space. Edwards reported that the total addressable market today is 8.6B (5B for TAVR, 1.5B for Surgical, 1B for TMTT, and 1.1B for Critical Care). They estimate that this market will grow to 20B by 2028 (10B TAVR, 2B Surgical, 5B TMTT, and 2B Critical Care. In 2021, their total sales was 5.23B, representing 60% of the market!! However, its always good to get future growth estimates from other sources outside the company as well. According to Fortune Business Insider, the heart valves market will balloon to $16B by 2026; a CAGR of 11.7%. According to the same source, Edwards is already the leader in the space.

It's important that a company be receiving revenue from a diversified portfolio of clients. Edwards is diversified not only in their products but also clientele. TAVR represents 65% of sales, TMTT at 12%, Surgical at 15% of sales, and critical care at 8%. Surgical and critical care's share of total sales have been decreasing, while TMTT's is rising, with TAVR remaining the same. Additionally, they sell both in the US (57% of sales) and internationally (mostly in Europe( 53% of international sales), Japan(23% of international sales), and rest of world. This is good that they are not too exposed to China. Not one of their customers represents 10% or more of their total sales, indicating that they have a diversified clientele. They sell mostly to hospitals and a few independent distributors.

One thing that is always good to look into is how the company performs during a recession to determine if they are recession proof. If not, how exposed are they if a recession occurs?
2020 Pandemic: sales still increased marginally, but rebounded strongly in 2021. This shows that even in a pandemic, their customers are still buying products.
2008/9 Recession: sales still increased albeit at a slower rate. 2007 sales grew by 11% while 2008 sales grew by 2.3%, and 2009 at 2.1%. It took until 2014 for sales to start seeing double digit growth again. However, sales have quadrupled since 2009! As the looming 2023 recession becomes more likely, it will be critically important to monitor the company's growth and compare it to the previous recessionary conditions.

Consequently, it's a good sign that a company is antifragile if they can raise prices into a recession and/or during high inflationary periods. While it is very difficult to track the true selling price of an Edwards valve, a good sign that management is not too concerned about inflation is the fact that word itself is not mentioned even once in the past five years of annual reports. If there was a pricing problem, management would most likely at least mention it.

Growth rates in order of importance: (are they moving in parallel? Why or why not?)

1. ROE - consistently over 20%, and recently has been increasing if you take out the litigation payment as well

2. EPS (took a dip in 2020 not because of covid, but because of a $400M litigation payment to Abbott labs regarding Mitraclip). Adjusted EPS still showed good growth
3. Sales - strong at over 10% per annum growth (except for 2020 b/c of covid) for the past ten years
4. Cash flow - cash flow has been increasing at a solid rate (15-20% for the past six years. This may not be reflected right away when you look at the cash flow statement, but taken into account the litigation payment and the 2017 income tax law, then cash flow has been consistent

**Income Statement:**

Gross margins have been hovering slightly above 70% for a decade. This business is extremely profitable and margins have actually been slightly increasing in the past five years. Operating expenses have been around 45% of total sales for a decade, and have actually been decreasing with time
As far as R&D, the Company spends roughly 17% of its total sales on R&D, which has increased in the past five years to build out the new TMTT business. The investment in this business will bring in billions of dollars of new revenues. Abbott averages 7%R&D spending compared to sales and Medtronic around the same. Edwards spends so much more on R&D because they have a business strategy of being first to market, and constantly staying one step ahead of their competition. This requires large R&D budgets, which can be costly to investors.
One a similar note, The company has been at 20% or higher earnings to sales ratio and climbing for the past decade or so (this is excluding 2020 because of the one-off litigation expense).

Net income has been consistently growing at 25.5% over the past adjusted 10 year period. This adjustment is taking into account one-off charges or income generated by unsustainable sources. For Edwards, the 2020 litigation payment of $400M to Abbott falls into this category. Another one-off change was the 2017 tax rate cut. Earnings per share (EPS) have been growing at the same rate. This is good to see as these two figures should grow together, indicating that management is not trying to buy back stock solely for the purpose of increasing EPS. On a similar note, basic and diluted EPS only differ by about 1.5% - 2.5%, indicating that employees of the company own that amount in options, and could exercise those options if they wanted. That difference is healthy and not too big to worry about.

TAXES: EW strategically pays a lower income tax rate than the federal rate of 21% by using California R&D credits, having a significant amount of their business abroad (whom have lower rates), tax benefits of employee share-based compensation, and by the changes in fair value of their liabilities. The real tax rate in 2021 was 11.7%, so they are legally avoiding paying too much in corporate income taxes, good to see for investors.

It's also important to note that Edwards does not pay a dividend to investors. Many times this is still acceptable IF AND ONLY IF management is using that money wisely to grow the business. EW reinvests all of its earnings back into the company. We are seeing a good use of the money by increasing ROIC, ROE, as well as increasing earnings at consistently increasing rates. I'd say management is doing a good job with the money they would've paid to investors as dividends.

**Balance Sheet:**

A company with a competitive advantage will often have 7ish years of surplus cash and be consistently adding to the surplus . Edwards is doing just this by growing its retained earnings at an astonishing rate around 25% - 30% per year. 2017 has a dramatic decrease in retained earnings because the company bought back 2.5B of stock.

Inventory should be slowly but consistently increasing showing they are producing more as a result of more sales . Edwards inventories have grown at 12% per year for the past 10 years. 15% in the past 5 years. This shows that business is only getting better. Inventories went down in 2021 because finished products decreased as a result of better sales

PPE: A company that is constantly increasing its PPE is usually in a fiercely competitive industry. A company with a strong moat doesn't have to update its PPE until it wears out (not often). It is also vital that well managed companies buy PPE, if any, with surplus cash rather than financing and going into debt. Edwards has been increasing its PPE at a CAGR of 18% for the past decade. This is not the best sign, but it can be attributed to the company investing heavily into its Irvine location with new buildings. It's hard to know if the same amount of building will continue into the future, could be a question for management. Additionally, these investments have been made through cash, which is a good sign.

Moving onto assets. The company has a fraction of the assets compared to its two biggest competitors (Medtronic and Abbott Labs). This could be attributed to the fact that their two competitors generate a lot more revenue than they do, however these companies are not solely focused on heart valves the way Edwards is. Edwards does not have to deal with as much depreciation as its competitors in this way. Staying hyper focused in one industry has its benefits. The company has averaged a return on assets (ROA) of 15% over the past five years. Medtronic is 5% and Abbott at 5.5%. Assets are still $8.5B, which is a significant amount of money if another company were raise and compete with Edwards. Not to mention the physician network moat that Edwards has built out, which would prevent a random company from competing with

Edwards. Additionally, they use a five year depreciation schedule, which is very fast. Most of their assets last longer than this, but they are no longer paying taxes on them (this is a smart accounting move).

Intangible assets: Company determines the fair value of its acquisitions by the discounted cash flows model. Intangible assets represent a small portion of total assets of the company (<10%) and most of it is from R&D and IP related activities. Edwards doesn't necessarily buy companies that are huge brand name companies, but buys them because of their IP as it relates to heart valves and critical care products.

Debt: Company took on long term debt twice in the past decade. In 2013 they took on $600M of notes at fixed rate at 2.8% and again in 2018 with another $600M also fixed rate but at slightly higher interest of 4.3%. The company is solvent enough to pay off the debt right now if they wanted to, and with a fixed rate so low, they can handle it.The most recent notes are due June 15th, 2028 and they plan on paying it off. This long term debt is far lower than the working capital of the business, book value, and can be paid off in less than one year's worth of earnings. It will not be an issue to the company.

Quick note on preferred stock here. Usually companies with strong moats don't have a lot, if any, preferred stock because it is expensive money for the company to offer. Currently, Edwards has given the board of directors the right to issue 50M shares of preferred stock in case of a liquidation event. The board has not yet issued these shares and therefore they have no material effect at the moment. However, 50M shares represents 7% of the total outstanding shares, and thus if they were to be issued, they would significantly dilute the common stock shareholders. That being said, a liquidation event is very unlikely.

The presence of treasury shares and a history of buying back shares is a good indicator of a moat. Both regular and treasury adjusted ROE are great and above 20% at the moment. Differences between them range anywhere from 5% to 20% depending on the year. Adjusted ROE removes the share buybacks from the calculation to see if management was using financial engineering to make EPS look good. A small deviation like this is okay, nothing to worry about here.

Let's move onto profitability:

- Abbott 5 year avg = 14% ROE
- Medtronic 5 year avg = 8.2% ROE
- Edwards 5 year avg  = 24% ROE

Enough said here.

**Cash Flow:**

The company splits it up into cash from financing as well as investing activities. For the past five years these two categories have consistently been negative. From the investing activities side, they are constantly investing in capital expenditures (averaging 6-8% of total sales, and 26% of total net income) as well as purchasing/selling investments - mostly purchasing, hence the consistently negative values. While cash from financing activities usually represents anywhere from 20%-45% compared to operating cash, that is because they are buying back stock using their cash. They bought back the most stock in the past five years in 2020, taking advantage of their decline in stock value. Although they also bought back a significant amount of shares in 2021 as well, even though the stock price was inflated. This could be a slight red flag, indicating management wanted to raise stock price to fulfill their option packages. It could also mean they simply want to own more of their own stock as any public company wants to. It's hard to really know the motivation behind stock buybacks sometimes. This is something to simply take note of and monitor as the years go by.

Operating cash is the most important cash flow generator, as it indicates the core business itself. Operating cash at Edwards is growing at an impressive average 21% per annum past ten years.

## ANALYSIS OF MANAGEMENT

A great CEO must be shareholder oriented, honest, and passionate. You want them to bleed the corporate colors. The CEO of Edwards is Mike Mussallem. He has been with Edwards since its break off from Baxter in 2000 - he's not going anywhere. Very focused on the future of this company. He owns about ~$500M of Edwards stock, which represents roughly 65% of his entire net worth, so it's safe to say he has a large vested interest in the performance of this company. Additionally, per the 2021 proxy statements: "Over the past five years, on average, 90% of the CEO's target total direct compensation has been performance-based, and 75% has been tied to the performance of Edwards' stock". This incentive plan is also not just based on stock price, but also on "key operating drivers" (KODs) which the company keeps secret but are vital towards maintaining their competitive advantage. Once again, per the 2021 proxy statement:

The Company's Annual Incentive Plan has historically consisted of three elements:

- the corporate financial measurement (based on underlying revenue growth, adjusted net income and adjusted free cash flow targets);

- the Key Operating Drivers (“KODs”) (quantifiable strategic milestones that include financial objectives and are tracked using a points system across our entire organization); and

- individual performance.

One potential red flag is that Mike does not write a letter to the shareholders. This could be because of a number of reasons. However, it's still possible to learn what kind of person he is through his involvement in podcasts, TV and media. A great CEO also must have incredible integrity. I think Mike Mussallen does have integrity because it shows in the way he talks about this business. He’s honest about when things are bad and good.

A good sign of potential future growth is if management and the CEO have a big audacious goal they are trying to achieve, and if this goal is unwavering through the thick and thin. It’s safe to say that Edwards does in fact have a strong, audacious goal which actually hasn’t changed since it’s inception in 2001: enhance patient outcomes for people with heart disease. In fact, as an employee myself I can say that there are signs and logos everywhere that say the corporate credo: “helping patients is our life's work, and life is now!”. By staying incredible patient focused, Edwards has been able to become the market leader in their space.

Numbers that reflect a good CEO:

- ROE 15% or better. Edwards at 28.8% in 2021
- ROIC 15% or better. Edwards at 25.5% in 2021
- Debt within 3 years of earning power. Edwards debt is within 0.7 years of earning power

Board of Directors is 8 people:

- **Martha Marsh**. Mostly experienced in health care roles and committees. Provides an overall perspective on health industry
    - Owns 52K shares (~$4.2M in value), 28K of which have been rewarded thus far for her directorship
- **Kieran Gallahue** → has been CEO of several medical device companies throughout his career. Offers financial and managerial perspectives
    - Owns about 97K shares (~$8M in value), half of which were rewarded thus far for his directorship since 2015

- **Leslie Heisz**→ mostly experience in corporate finance and banking. Helps with financial reporting and such
    - Owns 48K shares (~$4M in value), 25K of which have been rewarded thus far for her directorship
- **Paul LaViolette** → experience in medical device companies. Was COO of Boston Scientific for a bit as well as some startups. Provides perspective on strategy and innovation
    - Does not own stock or options
- **Steve Loranger** → Has held extensive executive positions and is an expert in manufacturing and operations, specifically in regulated industries
    - Owns 75K shares (~$6.5M in value), 15K of which were rewarded thus far for his directorship
- **Ramona Sequeira** → Only board member to not have current or previous directorships. Held several executive positions in pharmaceutical companies. Helps bring patient focused attitude.
    - Owns 3.3K shares ($264K in value), none of which have been rewarded thus far for her directorship
- **Nicholas Valeriani** → Worked at Johnson and Johnson in several director and executive roles. Provides experience in medical industry as well as strategy and innovation expertise
    - Owns 61K shares (~$5M in value), none of which were rewarded for his directorship

- Each director gets paid ~$300K per year

There are twelve corporate officers plus Mike Mussallem one for each business unit, in addition to some extra positions:

- TAVR: Larry Wood
- TMTT: Bernard Zovighian
- Surgical: Daveen Chopra
- Critical Care: Katie Szyman
- CSO: Todd Brinton
- CFO: Scott Ullem
- Huiman Wang: CVP of Japan, Asia, and Pacific
- Jean-Luc Lemercier: CVP of EMEA
- Don Bobo: CVP of corporate strategy and development

Almost every corporate officer has at least 25/30 years of experience in medical devices mostly in leadership roles

Executive compensation as said by the 2021 proxy statements: “Our executive compensation programs are designed to emphasize performance-based compensation, reward financial performance and the implementation of our corporate strategy and align the financial interests of our executives with those of our long-term stockholders.”

Over the past year, management and executives as a group have sold over a million shares of the company. Seems like management seemed pessimistic about 2022, for good reason. That being said, all executives combined own 1.3% of the company capitalization. Mike Mussallem himself owns 0.85% of that. This number is quite small and too much management ownership is far from an issue here.

It’s equally important to analyze the management as it is to analyze how employees feel about the company. As an employee myself I can say that employees of this company seem to be very happy and the culture here is very strong. Work life balance is emphasized heavily given the amount of athletic structures around, and team building exercises throughout the year

## ANALYSIS OF VALUATION

MOS (Margin of Safety) Price:
To determine total future growth rate, the averages of 10, 7, 5, and 3, years of sales, net earnings, and cash flow were calculated. Then, the average of these averages was calculated to determine one overall company growth rate.

- Keep in mind that the future growth rate of your company is fluid. It changes quarter to quarter, and year to year. Keep up with it to see how the story unfolds. Current growth rate averages are:
    - 25.25% based on earnings
    - 21.94% based on operating cash
    - 12.22% based on sales

TOTAL COMPANY GROWTH RATE (avg of all three averages) = 19.8% (let's call it 20%)

The trailing twelve months of EPS at the time of this writing is $2.44 per share. A minimally acceptable rate of return of 15% was chosen for the discount rate (recommended by Phil Town)

Using traditional discounted cash flows calculation, the true price of EW is $149 and the **50% off price is $75** per share. At the time of this writing, the stock price is trading at $85.95, meaning it’s very close to MOS price.

PAYBACK TIME:
Payback time is computed using the price one should pay to get paid back in 8 years of free cash flow growing at the average 10 year free cash flow growth rate. At the time of this writing, EPS of TTM is $2.44. Growing this at the historical free cash flow growth rate of 26%, and subsequently taking the summation of the future 8 years of free cash flow gives a **payback time price of $66 per share.**

OWNER EARNINGS:

Owner earnings was calculated using two methods: the Rule 1 method and the computation laid out by Ben Graham in the *Intelligent Investor* book.

OWNER EARNINGS RULE 1 WORKSHOP FROM 2021 FINANCIALS:

- \+ 1.7B (pretax income from income statement)
- \+ .13B (depreciation and amortization)
- \- .12B (income tax anticipated taking into account the tax paid for the one-off litigation with Abbott labs)
- \+ .2B (change in accounts payable from cash flow statement)
- \- .09B (change in receivable from cash flow statement)
- \- .33B - .14B (maintenance fund. Two numbers because one is capex reported and one is change is PPE reported. Both could be taken as maintenance)

= 1.51B - 1.68B Owner Earnings
10Cap price of 15.1B - 16.8B
**10Cap price per share: $24 - $27 per share**

OWNER EARNINGS FROM INTELLIGENT INVESTOR (2021 financials):

- \+ 1.59B (operating income)
- \+ .13B (depreciation and amortization)
- \- .12B (income tax anticipated taking into account the tax paid for the one-off litigation with Abbott labs)
- \- .1 (irregular income)
- \- .33B - .14 (capital expenditure)

= 1.17 - 1.36 B
10-cap price = 11.7B - 13.6B
**10-cap price per share: $19 - $22 per share**

Valuation Conclusion: Edwards is a growth focused company, the MOS and payback time prices are the most relevant way to calculate valuation, and thus it is my conclusion that Edwards is undervalued by the market by a margin of roughly 30-50%.